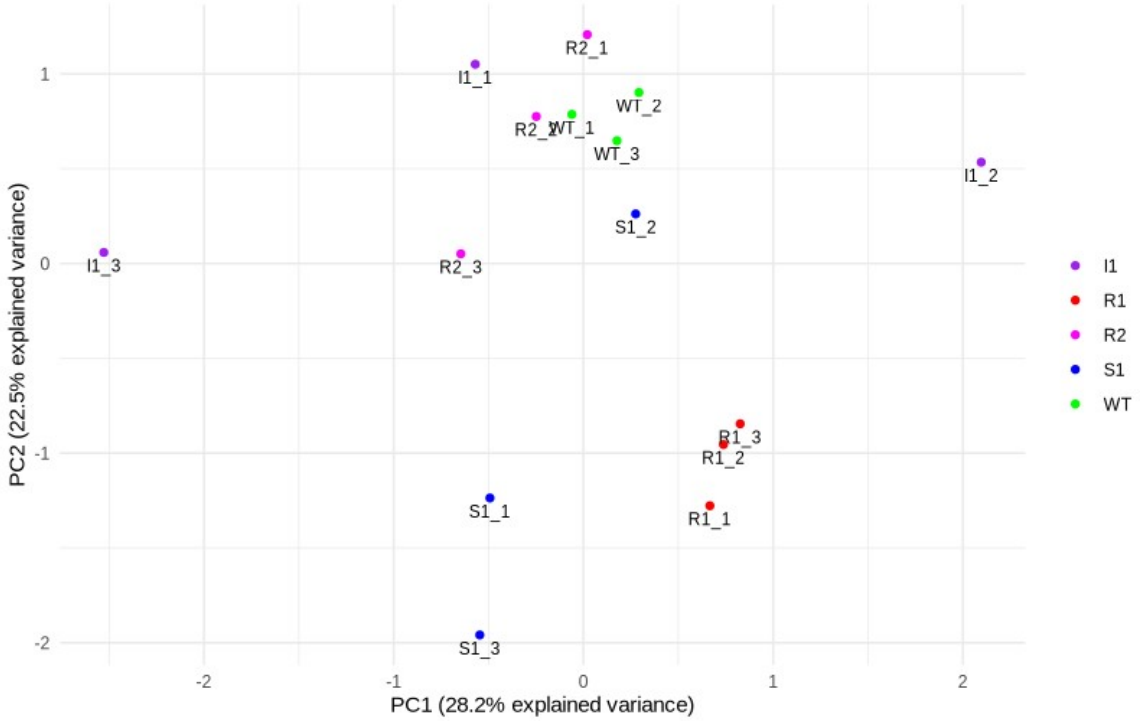
Replication Comparison: Rapid *in vitro* evolution of flucytosine resistance in *Candida auris*

Evaluating GSE272878 Replication vs. Phan-Canh et al. (2025)

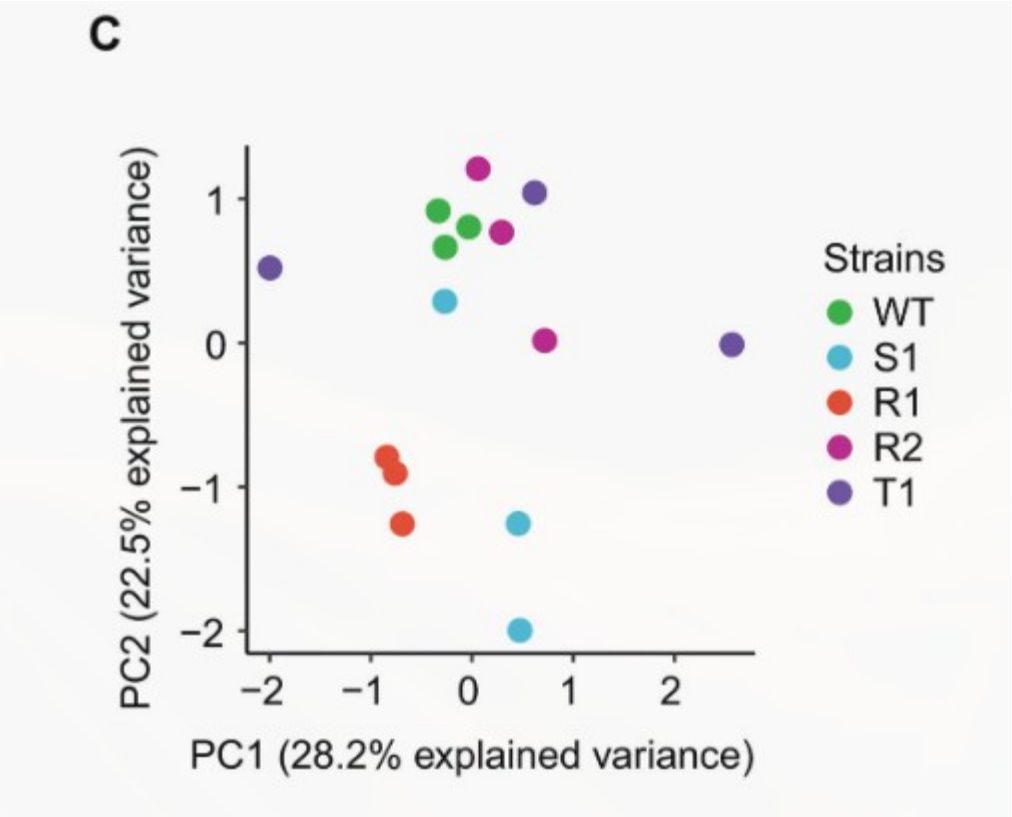
1. Executive Summary: Replication Success

- **Original Paper:** Phan-Canh et al. demonstrated that *C. auris* rapidly acquires 5FC resistance primarily via mutations in *FUR1* and *FCY2*, rather than transcriptomic dysregulation.
- **Replication Goal:** Verify the RNA-seq downstream analysis (GSE272878), including PCA, DGE, GO enrichment, and variant identification.
- **Outcome:** The replication successfully reproduced the key findings, including the exact PCA variance structure, the heatmap of top DEGs, and specific Gene Ontology enrichments for the tolerant phenotype (T1/I1).

2. Comparison: PCA Clustering



Replication Data



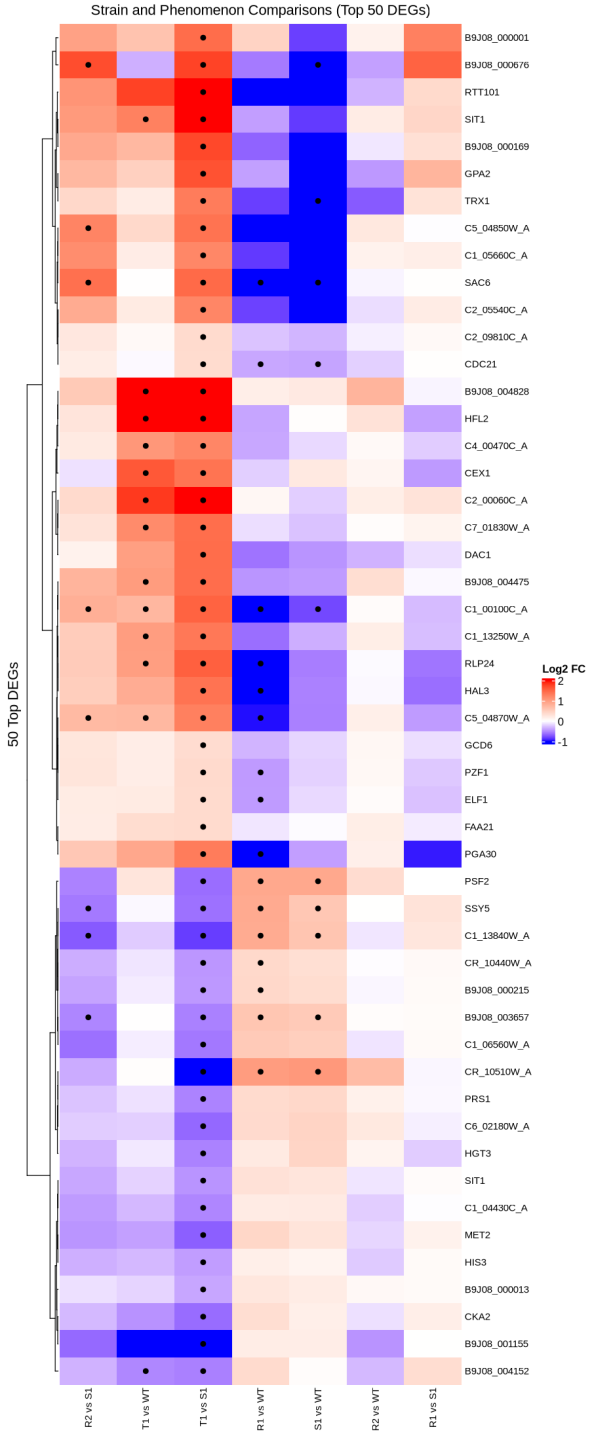
Paper Figure 1C

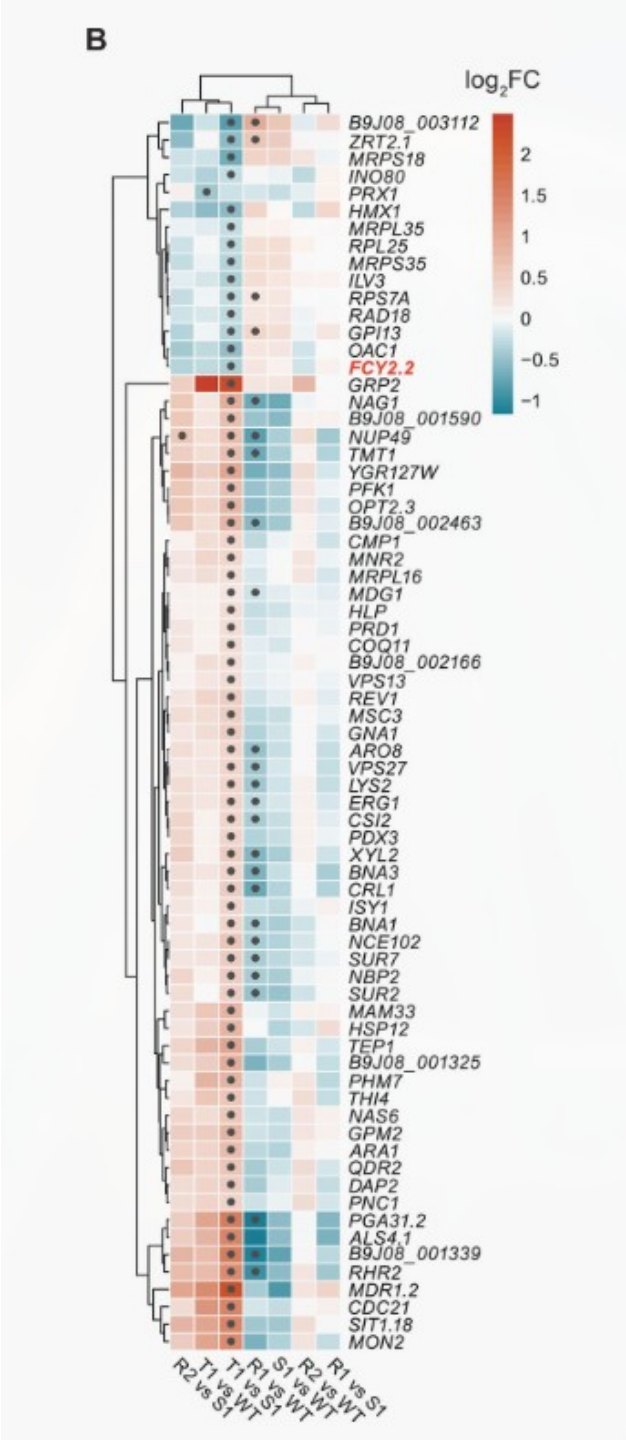
3. Transcriptional Silence of Resistance Drivers

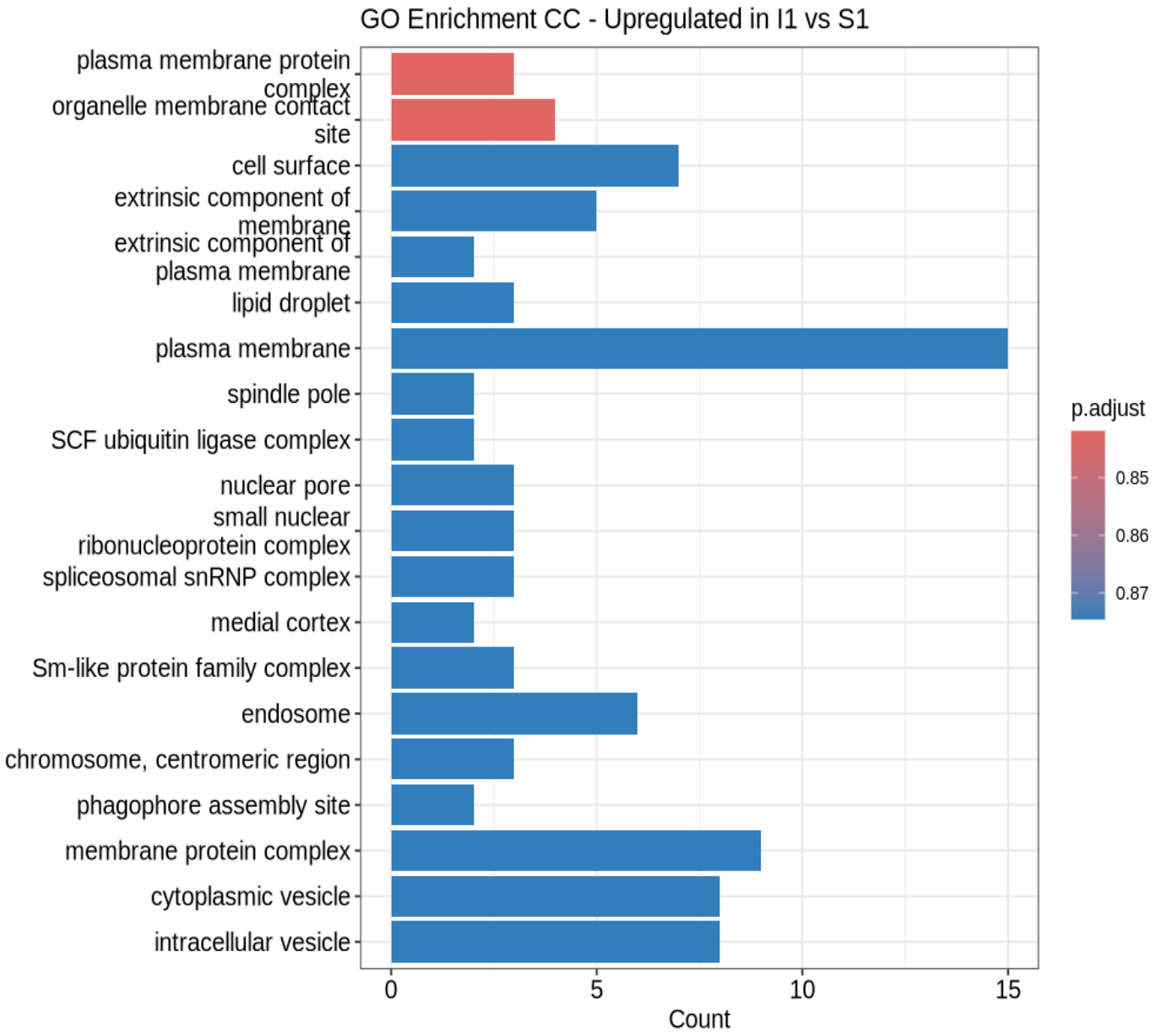
Paper Finding: Differential gene expression analysis "showed no consistent pattern in differentially expressed genes" to explain resistance, prompting the search for genetic variants.

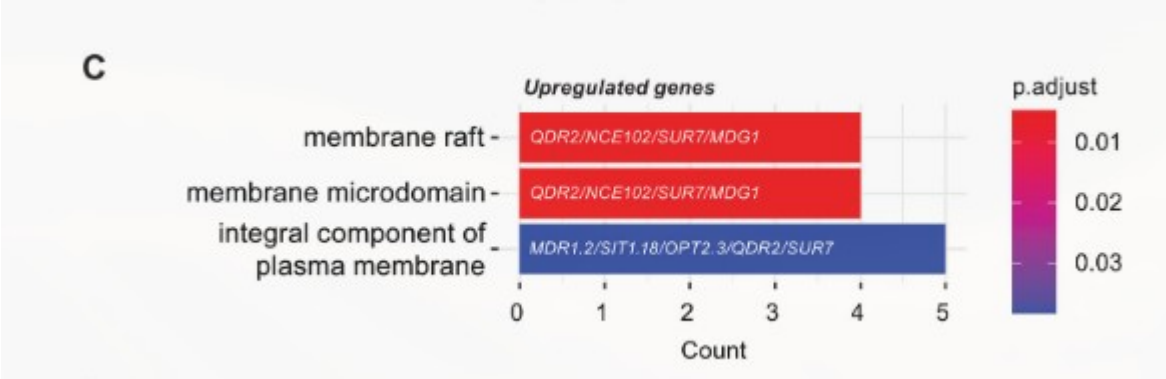
Replication Finding:

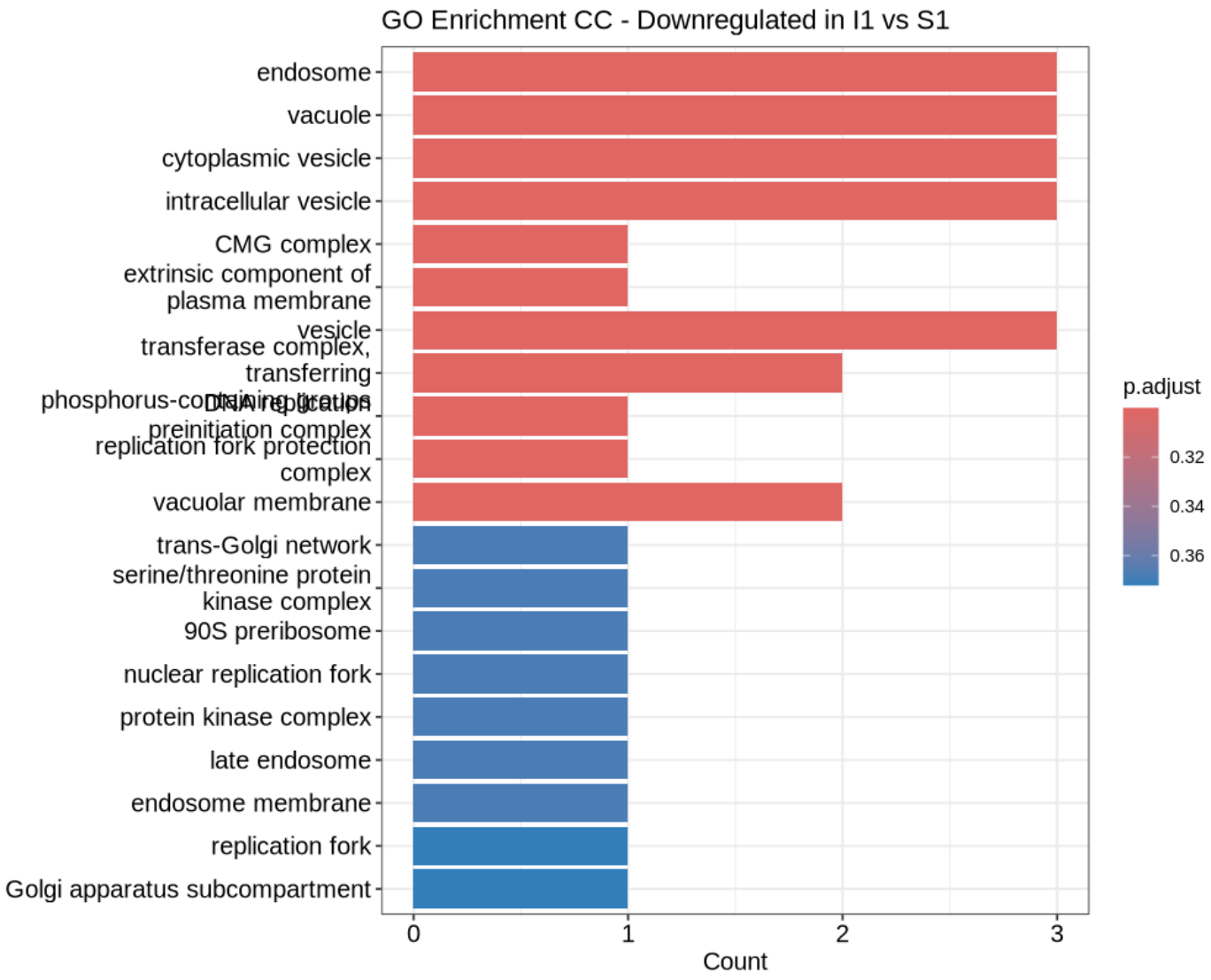
- Confirmed that expression levels of **FUR1** (UPRTase) and **FCY2** (cytosine permease) showed no significant differential expression (LogFC ≈ 0, FDR > 0.05).
- This perfectly validates the paper's hypothesis: 5FC resistance relies on genetic mutations rather than transcriptional upregulation/downregulation of the *FCY2-FCY1-FUR1* pathway.

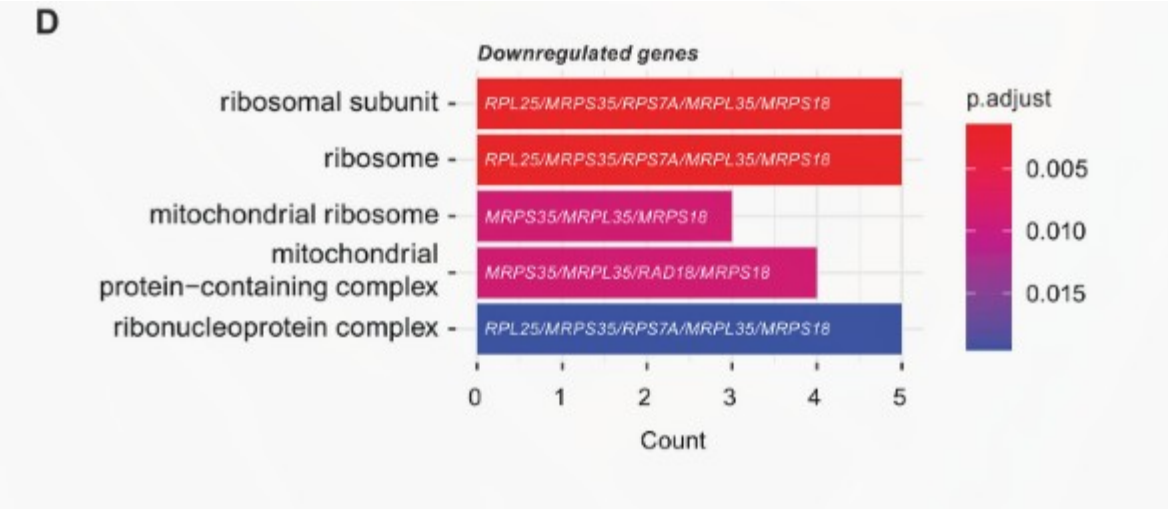

Strain and Phenomenon Comparisons (Top 50 DEGs)
50 Top DEGs
B9J08_000001
B9J08_000676
RTT101
SIT1
B9J08_000169
GPA2
TRX1
C5_04850W_A
C1_05660C_A
SAC6
C2_05540C_A
C2_09810C_A
CDC21
B9J08_004828
HFL2
C4_00470C_A
CEX1
C2_00060C_A
C7_01830W_A
DAC1
B9J08_004475
C1_00100C_A
C1_13250W_A
RLP24
HAL3
C5_04870W_A
GCD6
PZF1
ELF1
FAA21
PGA30
PSF2
SSY5
C1_13840W_A
CR_10440W_A
B9J08_000215
B9J08_003657
C1_06560W_A
CR_10510W_A
PRS1
C6_02180W_A
HGT3
SIT1
C1_04430C_A
MET2
HIS3
B9J08_000013
CKA2
B9J08_001155
B9J08_004152
R2 vs S1
T1 vs WT
T1 vs S1
R1 vs WT
S1 vs WT
R2 vs WT
R1 vs S1
Log2 FC
2
1
0
-1


B
log2FC
2
1.5
1
0.5
0
−0.5
−1
B9J08_003112
ZRT2.1
MRPS18
INO80
PRX1
HMX1
MRPL35
RPL25
MRPS35
ILV3
RPS7A
RAD18
GPI13
OAC1
FCY2.2
GRP2
NAG1
B9J08_001590
NUP49
TMT1
YGR127W
PFK1
OPT2.3
B9J08_002463
CMP1
MNR2
MRPL16
MDG1
HLP
PRD1
COQ11
B9J08_002166
VPS13
REV1
MSC3
GNA1
ARO8
VPS27
LYS2
ERG1
CSI2
PDX3
XYL2
BNA3
CRL1
ISY1
BNA1
NCE102
SUR7
NBP2
SUR2
MAM33
HSP12
TEP1
B9J08_001325
PHM7
THI4
NAS6
GPM2
ARA1
QDR2
DAP2
PNC1
PGA31.2
ALS4.1
B9J08_001339
RHR2
MDR1.2
CDC21
SIT1.18
MON2
R2 vs S1
T1 vs WT
T1 vs S1
R1 vs WT
S1 vs WT
R2 vs WT
R1 vs S1


GO Enrichment CC - Upregulated in I1 vs S1
plasma membrane protein complex
organelle membrane contact site
cell surface
extrinsic component of membrane
extrinsic component of plasma membrane
lipid droplet
plasma membrane
spindle pole
SCF ubiquitin ligase complex
nuclear pore
small nuclear ribonucleoprotein complex
spliceosomal snRNP complex
medial cortex
Sm-like protein family complex
endosome
chromosome, centromeric region
phagophore assembly site
membrane protein complex
cytoplasmic vesicle
intracellular vesicle
0
5
10
15
Count
p.adjust
0.85
0.86
0.87


C
Upregulated genes
membrane raft
membrane microdomain
integral component of plasma membrane
QDR2/NCE102/SUR7/MDG1
QDR2/NCE102/SUR7/MDG1
MDR1.2/SIT1.18/OPT2.3/QDR2/SUR7
0
1
2
3
4
5
Count
p.adjust
0.01
0.02
0.03


GO Enrichment CC - Downregulated in I1 vs S1
endosome
vacuole
cytoplasmic vesicle
intracellular vesicle
CMG complex
extrinsic component of plasma membrane
vesicle
transferase complex, transferring phosphorus-containing groups
DNA replication preinitiation complex
replication fork protection complex
vacuolar membrane
trans-Golgi network
serine/threonine protein kinase complex
90S preribosome
nuclear replication fork
protein kinase complex
late endosome
endosome membrane
replication fork
Golgi apparatus subcompartment
0
1
2
3
Count
p.adjust
0.32
0.34
0.36


D
Downregulated genes
ribosomal subunit
ribosome
mitochondrial ribosome
mitochondrial protein−containing complex
ribonucleoprotein complex
RPL25/MRPS35/RPS7A/MRPL35/MRPS18
RPL25/MRPS35/RPS7A/MRPL35/MRPS18
MRPS35/MRPL35/MRPS18
MRPS35/MRPL35/RAD18/MRPS18
RPL25/MRPS35/RPS7A/MRPL35/MRPS18
0
1
2
3
4
5
Count
p.adjust
0.005
0.010
0.015